

# Directed Acyclic Graphs (DAGs)

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A special class of directed graphs that is common in network science is those that do not have any cycles. They are referred to as directed acyclic graphs or DAGs. DAGs are common because they represent generalized hierarchy. And dependency networks in a generalized hierarchy and node may have more than one parent. An example of a dependency network is the prerequisite relations between college courses. We say that a directed network has a topological order, if we can run its nodes so that every adds points from a node of lower rank to a node of higher rank. This visualization shows the concept of a directed acyclic graph for a network with six nodes, and as you can see at the bottom, there is a topological order for this network, right? In which all the edges point from nodes at the left to nodes at the right. In other words, from nodes that are lower rank, to nodes that are higher rank. Here is a set of important facts about DAGs. First if a directed network has a topological order then it must be a DAG. Second, a DAG must include at least one source node. In other words a node with zero incoming edges. And third, if a graph is a DAG, then it must have a topological ordering, in other words directed network, has topological order if and only, it is a DAG. Here is how you can show the third property, that if a network is a DAG, then it must have a topological ordering. As we saw, if the network is a DAG, then it must have at least one source. So the first step of the algorithm is that we start from a source node  $s$ , in this example, as you can see, the DAG has two sources nodes A and G. Let's suppose for example that you start from node G. The next step of the algorithm is that you remove that source node, you remove  $g$  in this case, and also remove all the edges of that node. And so you decrement the indegree of all the nodes that that source was pointing to. The graph now is still a DAG. As you can see, after we have removed the source  $G$ , then you repeat the previous steps you choose a new source, in this case, source A, remove it, remove the edges of that node, and you

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repeat every time you identify the next node in the Topological Ordering. So here the source is B, you remove that. The next source is node E, you remove that and the process continues until you, you have basically removed all the nodes and all the nodes appear in the topological ordering.

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